

# Locked tortuosity metrics for WSS

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## Locked tortuosity and branching metrics for low-WSS work

Selection criteria for every pick below: (a) trustworthy and reproducible across different vessel trees, not an artifact of one dataset or one smoothing choice, and (b) most plausibly connectable, later, to low wall shear stress (WSS).

### Locked: three simple metrics

| Metric                              | Formula                                                                                                                           | Why it is locked                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | WSS evidence confidence                                                                                                                                                                                |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mean absolute curvature, $\kappa_a$ | $\kappa_a = \frac{1}{L} \sum_j \theta_j$ ,<br>$\theta_j = \arccos(u_j \cdot u_{j+1})$<br>over chord-resampled unit tangents $u_j$ | This project's own robustness test found $\kappa_a$ 's cohort-percentile ranking barely moved between $\sigma = 0$ and $\sigma = 1$ mm smoothing, the strongest cross-tree trustworthiness evidence of any candidate examined. It is also the only simple metric with a direct patient-cohort WSS finding: Kashyap/Gharleghi et al. 2022 (127-patient coronary CFD cohort) found average absolute curvature significantly correlated with percent low-TAWSS area in the LMCA and LAD. | Abstract/search-summary level. Nature.com login wall, PMC CAPTCHA, and a ResearchGate 403 blocked every full-text attempt, so the quote and any r/p-values are search-synthesis, not primary-verified. |

| Metric                                                       | Formula                                                                                                                                                                              | Why it is locked                                                                                                                                                                                                                                                                                                                                                                                                    | WSS evidence confidence                                                                                                                                                                                                                                                                                                                                                           |
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| RMS curvature, $\kappa_{rms}$                                | $\kappa_{rms} = \sqrt{\text{mean}_i(\kappa_i^2)}$<br>over the same<br>per-point curvature<br>sequence used for $\kappa_a$                                                            | Simple, and built from the same per-point curvature computation as $\kappa_a$ , so it is expected, though not independently tested, to inherit comparable smoothing stability. It sat in the same head-to-head test as $\kappa_a$ in the one identified patient-cohort WSS study, giving it the best-evidenced pedigree of any metric not already locked as the primary pick.                                       | Weaker than $\kappa_a$ , and an explicit open point: Kashyap et al. 2022 tested RMS curvature against percent low-TAWSS area alongside $\kappa_a$ , but the notes gathered for this task do not confirm whether RMS curvature was itself statistically significant, only that average absolute curvature was the metric the paper recommended. Treat as plausible, not confirmed. |
| Horton-Strahler bifurcation ratio, $R_B \approx N_i/N_{i+1}$ | Recursive branch ordering (a tributary-free segment is order 1; two merging order- $i$ segments produce order $i + 1$ ); $R_B$ is the segment-count ratio between consecutive orders | The only branching-topology pick, so it covers a genuinely different axis than the two curvature-family picks above. Its trustworthiness case rests on generality rather than a project-specific test: the same construction has held up for nearly 80 years across river networks, neuronal dendritic trees, and vascular trees, and being a discrete count ratio it carries no smoothing-scale dependence at all. | No direct WSS study was found in the notes for this metric. Its link to WSS is a plausible-but-unverified extension only: branch order governs daughter-vessel caliber, and caliber together with bifurcation geometry is exactly what the EuroIntervention review below ties directly to local endothelial shear stress. This is inference, not a finding.                       |

**Locked: two advanced extraction methods**

| Method            | Definition                                                                                                                                                                                                                         | Why it is locked                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | WSS evidence confidence                                                                                                                                                                                                                                                                                                            |
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| Bifurcation angle | $\theta_{bif} = \arccos(d_1 \cdot d_2)$ , with $d_1, d_2$ the two daughter-branch tangent directions at a bifurcation, extracted via a robust local fit (this project’s own fitcore3 method) rather than a naive three-point angle | Strongest trustworthiness AND strongest WSS case of every candidate examined across all three research passes, both primary-source-verified. fitcore3 was shown stable at 78.0 to 80.8 degrees across different smoothing choices on this project’s own data. On the WSS side, three primary sources were read in full: a EuroIntervention review states directly that wider bifurcation angle and diameter lower local endothelial shear stress; Sun et al. (30 patients, CCTA-derived CFD) found an 80-degree angle threshold predicting significant stenosis at 100 percent sensitivity; and a deep-learning surrogate trained on 1,800 idealized bifurcation geometries reproduced the same wider-angle-to-lower-WSS/higher-OSI relationship. | Primary-source-verified, the only fully-verified WSS claim among all five locked picks. One caveat carries over: the angle-WSS direction reverses in already-stenosed vessels (wider angle correlates with higher local WSS at the lesion itself), a different regime from the pre-disease geometric-predisposition studies above. |

| Method            | Definition                                                                                                                                                                                    | Why it is locked                                                                                                                                                                                                                                                                                                                                              | WSS evidence confidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| Torsion, $\tau_a$ | $\tau_a = \frac{1}{L} \sum_j \phi_j$ , $\phi_j$ the turning angle of the discrete binormal-normal frame between consecutive chord-resampled points, the Frenet-Serret companion to $\kappa_a$ | Chosen over the two other advanced candidates with good trustworthiness credentials, the Richardson-Mandelbrot scale-robust curvature exponent and the Topological Morphology Descriptor, specifically because those two have no direct WSS evidence anywhere in the notes, while torsion has at least a real, if contested, empirical link via helical flow. | Abstract/search-summary level only, and the directionality is genuinely contested. Gallo et al. report torsion correlates with helical-flow intensity ( $r = 0.64$ ) and helical-flow intensity anti-correlates with unfavorable WSS ( $r = -0.91$ ), but a 2025 39-patient ASOCA CFD study found helicity is a “double-edged sword”: it reduces low-WSS area but simultaneously increases high-WSS/oscillatory area linked to plaque vulnerability. Neither source’s full text could be verified past the abstract. Torsion’s own cross-tree trustworthiness is also the weakest of the five locks: unlike $\kappa_a$ and bifurcation angle, no project-specific robustness test exists yet; the Frenet-Serret torsion companion is currently a design proposal, not a validated, run pipeline. |

#### Also considered, not chosen

- **TI/DM (arc-chord tortuosity index)**: classic and the simplest of all, but showed no significant correlation with low-TAWSS area in the one patient-cohort study found (Kashyap et al. 2022,  $p > 0.05$ ), and is confounded by vessel length at  $\rho = 0.7$  to  $0.84$ .

- **Sinuosity-index banding:** fixed thresholds applied on top of TI/DM, so it inherits TI/DM's null WSS result and length confound.
- **Inflection Count Metric (ICM):** simple and well-defined, but no WSS study of any kind surfaced for it.
- **Peak / 95th-percentile local curvature:** the one relevant source (Johnston & Johnston 2007) found high/extreme curvature only weakly correlated with WSS extremes, weaker than reciprocal radius, an argument against rather than for this pick.
- **Bullitt Triangular Index:** a genuinely distinct local straightness measure, but no WSS evidence was found for it.
- **Average squared-derivative curvature:** tested in the same Kashyap et al. 2022 cohort as  $\kappa_a$  and RMS curvature, but its exact formula could not be confirmed and its own significance against TAWSS is unresolved, a double uncertainty that ruled it out.
- **Raw chord  $D$  and raw arc length  $L$  as standalone descriptors:** useful as confound controls, not tortuosity or branching descriptors in their own right; no WSS link was proposed for them anywhere.
- **Richardson-Mandelbrot scale-robust curvature exponent:** excellent trustworthiness case, purpose-built to be robust to the smoothing-scale choice that plagues every other curvature metric, but no WSS study of any kind was found for it.
- **Topological Morphology Descriptor (TMD):** strong, primary-verified trustworthiness case (proven strictly more informative than Sholl analysis via persistent homology), but likewise no WSS evidence exists; it is a whole-tree branching descriptor with no tested biomechanical link.
- **WAL/WAC whole-tree aggregation formulas:** a validated way to collapse many vessel segments into one number, but an aggregation method for other metrics, not itself a metric with WSS evidence.
- **Non-planarity / out-of-plane angle** (standalone, distinct from torsion): the notes' own ranking places this as the weakest of the three torsion-family candidates, evidenced only qualitatively (non-planar vs. planar CFD comparisons), never as a scalar predictor in a statistical model.